

$$\begin{aligned}
C_{quqd}^{(8)} i1_{-} i2_{-} i3_{-} i4_{-} \rightarrow & \frac{1}{16\pi^2} \left(\frac{1}{18} s_\gamma c_\gamma \frac{1}{m_\Phi^2} \left(9 s_\gamma^2 \overline{y_d^{pr}} y_d^{i1i4} y_d^{i3r} y_u^{pi2} - \right. \right. \\
& 36 g_3^2 y_d^{i3i4} y_u^{i1i2} - y_u^{i3i2} \left(y_d^{i1i4} (5 g_1^2 + 27 g_2^2 + 24 g_3^2) - 9 c_\gamma^2 y_d^{pi4} \overline{y_u^{pr}} y_u^{i1r} \right) \Big) + \\
& \frac{1}{9} s_\gamma c_\gamma \left(-36 g_3^2 y_d^{i3i4} y_u^{i1i2} - y_d^{i1i4} y_u^{i3i2} (5 g_1^2 + 27 g_2^2 + 24 g_3^2) \right) LF_{1,2}[m_\Phi] + \\
& \frac{2}{9} m_1 \tilde{\mu} g_1^2 y_d^{i1i4} y_u^{i3i2} LF_{1,1,1,1,0}[m_1, m_{\tilde{d}}^{i4}, m_{\tilde{q}}^{i3}, \tilde{\mu}] + \\
& \frac{8}{9} m_1 \tilde{\mu} g_1^2 y_d^{i1i4} y_u^{i3i2} LF_{1,1,1,1,0}[m_1, m_{\tilde{d}}^{i4}, m_{\tilde{u}}^{i2}, \tilde{\mu}] - \\
& \frac{1}{9} m_1 \tilde{\mu} g_1^2 y_d^{i1i4} y_u^{i3i2} LF_{1,1,1,1,0}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i3}, \tilde{\mu}] - \\
& \frac{4}{9} m_1 \tilde{\mu} g_1^2 y_d^{i1i4} y_u^{i3i2} LF_{1,1,1,1,0}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{i2}, \tilde{\mu}] + \\
& 3 m_2 \tilde{\mu} g_2^2 y_d^{i1i4} y_u^{i3i2} LF_{1,1,1,1,0}[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i3}, \tilde{\mu}] + \\
& \frac{2}{3} m_3 \tilde{\mu} g_3^2 y_d^{i1i4} y_u^{i3i2} LF_{1,1,1,1,0}[m_3, m_{\tilde{d}}^{i4}, m_{\tilde{q}}^{i3}, \tilde{\mu}] + \\
& \frac{2}{3} m_3 \tilde{\mu} g_3^2 (3 y_d^{i3i4} y_u^{i1i2} + y_d^{i1i4} y_u^{i3i2}) LF_{1,1,1,1,0}[m_3, m_{\tilde{d}}^{i4}, m_{\tilde{u}}^{i2}, \tilde{\mu}] + \\
& \frac{8}{3} m_3 \tilde{\mu} g_3^2 y_d^{i3i4} y_u^{i1i2} LF_{1,1,1,1,0}[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i3}, \tilde{\mu}] + \\
& \frac{2}{3} m_3 \tilde{\mu} g_3^2 y_d^{i1i4} y_u^{i3i2} LF_{1,1,1,1,0}[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{i2}, \tilde{\mu}] \Big)
\end{aligned}$$